Giacomo Picardi

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ICM-CSIC, Barcelona, Spain

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EDUCATION

The BioRobotics Institute,
Scuola Superiore Sant'Anna, Pisa, Italy
Ph.D. in Biorobotics, December 2019 — Score: 100/100 with Honors
Thesis: A bioinspired approach to underwater legged robotics

Università di Pisa, Italy
M.Sc. in Robotics and Automation Engineering, April 2016 — Score: 110/110 with Honors
Thesis: Design of a model based adaptive controller for the hovering manoeuvre of a small civil helicopter

Università di Pisa, Italy
B.Sc. in Computer Engineering, December 2013 — Score 110/110 with Honors
Thesis: Development of first-person camera for a quadcopter simulator based on Google Earth API

RESEARCH EXPERIENCE

Post-doctoral researcher — Instituto de Ciencias del Mar (ICM-CSIC), Barcelona, Spain
January 2023 – Present
Mentor: Dr. Jacopo Aguzzi

Visiting scholar — Universitat Politècnica de Catalunya (UPC), Barcelona, Spain
September 2023 – September 2024
Mentor: Dr. Joaquin del Rio

Post-doctoral researcher — Scuola Superiore Sant'Anna (SSSA), Pisa, Italy
February 2021 – January 2023
Mentor: Prof. Matteo Cianchetti

Post-doctoral researcher — Scuola Superiore Sant'Anna (SSSA), Pisa, Italy
January 2020 – February 2021
Mentor: Prof. Marcello Calisti

Visiting scholar — NASA Jet Propulsion Laboratory, Pasadena (CA), USA
April 2019 – September 2019
Mentor: Dr. Kalind Carpenter

Ph.D. candidate — Scuola Superiore Sant'Anna (SSSA), Pisa, Italy
October 2016 – December 2019
Supervisor: Prof. Cecilia Laschi

Master thesis — Max Planck Institute for Biological Cybernetics, Tübingen, Germany
September 2015 – March 2016
Supervisor: Prof. Dr. Heinrich Bülthoff

Bachelor thesis — Università di Pisa, Italy
September 2013 – December 2013
Supervisor: Prof. Lorenzo Pollini

FELLOWSHIPS, GRANTS AND DISTINCTIONS

Ramon y Cajal — September 2027 -
(2025 – ID: RYC2025-055672-I)

La Caixa Junior Leader fellowship — December 2026 – December 2029
(Application ID: 161835)
Instituto de Ciencias del Mar (ICM-CSIC)

CSIC AGAIN (ID: AGAIN26010) — November 2026
ERC StG-2026 proposal B-BRUI, interview stage, evaluated B. Funding: 10,000 €

Overseas Visiting Scholar program — October 2026 – November 2026
Harbin Institute of Technology, Shenzhen

Marie Curie post-doctoral fellowship — September 2023 – August 2025
(Grant agreement number 101061354)
Instituto de Ciencias del Mar (ICM-CSIC)

Beatriu de Pinós post-doctoral fellowship — January 2023 – August 2023 (terminated early to take up Marie Curie Postdoctoral Fellowship)
(Grant agreement number 2021 BP 00039)
Instituto de Ciencias del Mar (ICM-CSIC)

International mobility scholarship — April 2019 – September 2019
NASA Jet Propulsion Laboratory

Ph.D. scholarship — October 2016 – September 2019
Scuola Superiore Sant'Anna

Erasmus+ mobility scholarship — September 2015 – March 2016
Max Planck Institute for Biological Cybernetics

Erasmus mundi mobility scholarship — September 2012 – June 2013
University College of London (UCL)

PUBLICATIONS

JOURNALS

1. Aguzzi J., Chatzidouros E., Buscaino G., Chatzievangelou D., **Picardi G**., …& Papaelias M. (2026), A Policy-Driven Design of Energy-Autonomous Intelligent Observatories for Marine Biodiversity Monitoring, Engineering, https://doi.org/10.1016/j.eng.2026.07.007.
2. Aguirregomezcorta, J., Aguzzi, J. & **Picardi, G.** (2026). A Review and Case Study on Underwater Legged Robot Simulation. Accepted for publication in Elsevier Ocean Engineering.
3. **Picardi, G.**, Iacoponi, S., Carandell, M., Aguirregomezcorta, J., Chellapurath, M., del Rio, J., ... & Aguzzi, I. (2025). An Open Toolkit for Underwater Field Robotics. arXiv preprint arXiv:2512.15597.
4. Ortenzi, L., Aguzzi, J., Chatzievangelou, D., Nemmi, E. N., Ferrari, M., Masmitja, I., ... & Costa, C. (2025). Generative artificial intelligence and marine ecological monitoring. Environmental Modelling & Software, 106789.
5. Aguzzi, J., Chatzidouros, E., Chatzievangelou, D., Clavel-Henry, M., Flögel, S., Bahamon, N., ... & Doyle, J. (2025). A digital-twin strategy using robots for marine ecosystem monitoring. Ecological Informatics, 103409.
6. Morane, C. H., Sascha, F., Mihai, T. D., Gerard, L. T., Nixon, B., **Picardi, G**., ... & Jacopo, A. (2025). A roadmap for designing a virtual interface to explore a digital twin of the oceans

for marine ecosystems monitoring. Ecological Informatics, 103483.

7. **Picardi, G**., Astolfi, A., Seemakurthy, K., Hurst, B., Piana, E., Bosilj, P., & Calisti, M. (2025). Visual servoing of an underwater robotics arm for automatic sorting of crustaceans. Smart Agricultural Technology, 12, 101168.
8. Aguzzi, J., Cuadros, J., Dartnell, L., Costa, C., Violino, S., Canfora, L., ... & Foing, B. (2024). Marine Science Can Contribute to the Search for Extra-Terrestrial Life. Life, 14(6), 676.
9. Aguzzi, J., Thomsen, L., Flögel, S., Robinson, N. J., **Picardi, G**., Chatzievangelou, D., ... & Danovaro, R. (2024). New technologies for monitoring and upscaling marine ecosystem restoration in deep-sea environments. Engineering, 34, 195-211.
10. Astolfi, A., Chellapurath, M., **Picardi, G**., Capriotti, M., Mladinich, K., Laschi, C., ... & Calisti, M. (2024). Marine sediment sampling with an underwater legged robot: A user-driven sampling approach for microplastic analysis. IEEE Robotics & Automation Magazine, 31(1), 62-71.
11. **Picardi, G**., Astolfi, A., Chatzievangelou, D., Aguzzi, J., & Calisti, M. (2023). Underwater legged robotics: Review and perspectives. Bioinspiration & Biomimetics, 18(3), 031001.
12. Falahzadeh, A., Toma, D. M., Francescangeli, M., Chatzievangelou, D., Nogueras, M., Martínez, E., ... & del Río, J. (2023). A new coastal crawler prototype to expand the ecological monitoring radius of OBSEA cabled observatory. Journal of Marine Science and Engineering, 11(4), 857.
13. **Picardi, G**., De Luca, M., Chimienti, G., Cianchetti, M., & Calisti, M. (2023). User-driven design and development of an underwater soft gripper for biological sampling and litter collection. Journal of Marine Science and Engineering, 11(4), 771.
14. Aguzzi, J., Flögel, S., Marini, S., Thomsen, L., Albiez, J., Weiss, P., ... & Danovaro, R. (2022). Developing technological synergies between deep-sea and space research. Elem Sci Anth, 10(1), 00064.
15. Aguzzi, J., Chatzievangelou, D., Robinson, N. J., Bahamon, N., Berry, A., Carreras, M., ... & Doyle, J. (2022). Advancing fishery-independent stock assessments for the Norway lobster (Nephrops norvegicus) with new monitoring technologies. Frontiers in marine science, 9, 969071.
16. Astolfi, A., **Picardi, G**., & Calisti, M. (2021). Multilegged underwater running with articulated legs. IEEE Transactions on Robotics, 38(3), 1841-1855.
17. Chellapurath, M., Walker, K. L., Donato, E., **Picardi, G**., Stefanni, S., Laschi, C., ... & Calisti, M. (2021). Analysis of station keeping performance of an underwater legged robot. IEEE/ASME Transactions on Mechatronics, 27(5), 3730-3741.
18. George Thuruthel, T., **Picardi, G**., Iida, F., Laschi, C., & Calisti, M. (2021). Learning to stop: a unifying principle for legged locomotion in varying environments. Royal Society Open Science, 8(4), 210223.
19. **Picardi, G**., Hauser, H., Laschi, C., & Calisti, M. (2021). Morphologically induced stability on an underwater legged robot with a deformable body. The International Journal of Robotics Research, 40(1), 435-448.
20. **Picardi, G**., Borrelli, C., Sarti, A., Chimienti, G., & Calisti, M. (2020). A minimal metric for the characterization of acoustic noise emitted by underwater vehicles. Sensors, 20(22), 6644.
21. **Picardi, G**., Chellapurath, M., Iacoponi, S., Stefanni, S., Laschi, C., & Calisti, M. (2020). Bioinspired underwater legged robot for seabed exploration with low environmental disturbance. Science Robotics, 5(42), eaaz1012.
22. **Picardi, G**., Laschi, C., & Calisti, M. (2018). Model-based open loop control of a multigait legged underwater robot. Mechatronics, 55, 162-170.
23. Calisti, M., **Picardi, G**., & Laschi, C. (2017). Fundamentals of soft robot locomotion. Journal of The Royal Society Interface, 14(130), 20170101.

CONFERENCES

1. Masmitja, I, Mario Martin, Raul Bardaji, ... Nathan Robinson TECTUGA: using innovative animal-borne cameras and autonomous vehicles for ecological assessments of sea turtles

and their habitats. In IEEE Oceans 2026, Monterrey, USA.

2. Costin Boiciuc, **Giacomo Picardi**, Joaquin del Rio, Matias Carandell, Daniel Toma, Rolling Locomotion for Shallow-Water Benthic Exploration Using a Compact Spherical Underwater Robot. In IEEE Oceans 2026, Sanya, China.
3. Aguirregomezcorta, J., Aguzzi, J., & **Picardi, G**. (2025). Simulación de Robots Submarinos con Patas: un enfoque basado en ROS 2. In ROSCon España 2025.
4. Aguirregomezcorta, J., Aguzzi, J., & **Picardi, G.** (2025). Simulating Underwater Legged Robotics with a ROS2-based approach. In OCEANTECH workshop at 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
5. Papaelias, M., Gee, M., Bratchetta, V., Zhang, Z., Hayati, F., Roshanmanesh, S., … & Chronakis, A. (2025). Design and demonstration of a novel wave energy converter for promoting marine habitat restoration. In Proceedings of the 15th TSME International Conference on Mechanical Engineering.
6. **Picardi, G**., Chatzievangelou, D., Clavel-Henry, M., Francescangeli, M., Masmitja, I., Quevedo, J., … & Aguzzi, J. (2025). From mobile and fixed monitoring platforms to the Digital Twin of the Ocean: Synergies and impacts of the three European projects DIGI4ECO, SUN-BIO, and MERLIN. In MarTech 2025 – Marine Technology Workshop (Pasaia, Gipuzkoa, Spain, May 28–29).
7. Papaelias, M., Hayati, F., Roshanmanesh, S., Fernando, G. F., Gee, M., Zhang, Z., ... & Grigoropoulos, G. (2024). Sunbio: Towards Widespread Implementation of Wave Energy Converters. Institute of Electrical and Electronics Engineers.
8. Chatzievangelou, D., Aguzzi, J., Bahamon, N., Biel Cabanelas, M., Cecchini, F. M., Clavel-Henry, M., ... & REDRESS, D. (2024). Marine biodiversity monitoring in the age of intelligent robots: Needs, challenges and insights.
9. Gee, M., Aguzzi, J., **Picardi, G**., Masmitja Rusiñol, I., Chatzievangelou, D., Bahamon Rivera, N., ... & Papaelias, M. (2024). Technical challenges in achieving widespread implementation of wave energy converter systems and future directions. In Proceedings of the 77th IIW Annual Assembly and International Conference, Rhodes, Greece, July 7-12, 2024. Welding Greek Institute (WGI).
10. **Picardi, G**., Del Rio, J., & Aguzzi, J. (2024). Bioinspired Legged Underwater robots for Ecological monitoring and Exploration. In UNESCO Ocean Decade Conference, 10th-12th April 2024, Barcelona, Spain.
11. Francescangeli, M., Chatzievangelou, D., Llorach-Tó, G., Castellicchio, A., Río, J. D., Toma, D. M., ... & Aguzzi, J. (2024). The OBSEA demo-site within the DIG4ECO Project.
12. **Picardi, G**., Astolfi, A., & Calisti, M. (2024). Seabed intervention with an underwater legged robot. In 2024 IEEE International Conference on Robotics and Automation (ICRA) (pp. 6247-6253). IEEE.
13. **Picardi, G**., Carandell, M., Nogueras, M., Toma, D., Río Fernández, J. D., & Aguzzi, J. (2024). An open-source control architecture for teleoperated underwater legged robots. In MarTech 2024 – Marine Technology Workshop (Palma, Mallorca, Spain, June 6-7).
14. Pagliarani, N., **Picardi, G**., Pathan, R., Uccello, A., Grogan, H., & Cianchetti, M. (2023). Towards a bioinspired soft robotic gripper for gentle manipulation of mushrooms. In 2023 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor) (pp. 170-175). IEEE.
15. Pagliarani, N., **Picardi, G**., Zaidi, S. S. Z., & Cianchetti, M. (2023). Variable kinematics enabled by layer jamming transition in a soft bending actuator. In 2023 IEEE International Conference on Soft Robotics (RoboSoft) (pp. 1-7). IEEE.
16. Donato, E., **Picardi, G.,** & Calisti, M. (2021). Statics optimization of a hexapedal robot modelled as a Stewart platform. In Annual Conference Towards Autonomous Robotic Systems (pp. 370-380). Cham: Springer International Publishing.
17. Mirimin, L. et al. (2021). Using cabled observatories as in situ platforms for advancing novel non-invasive marine biodiversity monitoring approaches. In: ASLO 2021 Aquatic Sciences Meeting. Association for the Sciences of Limnology and Oceanography.
18. **Picardi, G**., Lovecchio, R., & Calisti, M. (2021). Towards autonomous area inspection with a bio-inspired underwater legged robot. In 2021 IEEE/RSJ International Conference on

Intelligent Robots and Systems (IROS) (pp. 930-935). IEEE.

19. Calisti, M., **Picardi, G**., & Laschi C. (2019). Bioinspired robots for seabed exploration. In 2019 IRIM 3D Event. Italian Institute of Robotics and Intelligent Machines (IRIM). ISBN: 9788894580501
20. **Picardi, G**., Chellapurath, M., Iacoponi, S., Laschi, C., & Calisti, M. (2019). Surveying and cleaning plastic pollution in the sediment: SILVER+ approach. In OCEANS 2019-Marseille (pp. 1-8). IEEE.
21. Iacoponi, S., **Picardi, G**., Chellapurath, M., Calisti, M., & Cecilia, L. (2018). Underwater soft jet propulsion based on a hoberman mechanism. In 2018 IEEE International Conference on Soft Robotics (RoboSoft) (pp. 449-454). IEEE.
22. **Picardi, G**., Geluardi, S., Olivari, M., Pollini, L., Innocenti, M., & Heinrich, H. B. (2016, April). L1-based model following control of an identified helicopter model in hover. In 72nd American Helicopter Society International Annual Forum 2016: Leveraging Emerging Technologies for Future Capabilities (pp. 1770-1777). American Helicopter Society.

PROJECTS

1. **BEN-FLOW –** Breaking the underwater intervention paradigm. Dec 2026 – Dec 2029. PI: Dr. Giacomo Picardi (ICM-CSIC). Funding: 320,000 €, la Caixa Foundation. Role: PI and grantee.

2. **AI4SEA -** Artificial Intelligence for Sea Ecological Advanced monitoring and restoration. Dec 2025 – Nov 2028. PI: Dr. Nixon Bahamon and Dr. Jacopo Aguzzi (ICM-CSIC). Funding: 1,465,000 €, Ministerio de Ciencias, Innovación y Universidades (G.A. IA2025-163346-C44). Role: Research and Work Package Leader (WP5 - Ecological Data Integration and Digital Twin Development for Monitoring and Mapping).

3. **TECTUGA** – Sentinels of the Sea: using innovative animal-borne cameras and autonomous vehicles for ecological assessments of sea turtles and their habitats. Dec 2025 – Nov 2028. PI: Dr. Nathan Robinson, Dr. Ivan Masmitja. Funding: 228.584,00 €, Ministerio de Ciencia e Innovación (PID2024-161772OA-I00). Role: Researcher.

4. **MERLIN** – Demonstration of long-endurance intelligent multi-purpose autonomous vehicles for marine applications. Nov 2024 – Oct 2028. PI: Dr. Papaelias (UoB). Funding: 7,946,786.25 €, European Union (G.A. 101189796). Role: Task Leader (T2.1 – Coordinated mission definition and operational requirements).

5. **SUNBIO** – Sustainable Nature and Inclusive Offshore Energy with Parallel Biodiversity Flourishing, Protection and Monitoring. Jun 2024 – May 2028. PI: Dr. Aguzzi (ICM-CSIC). Funding: 2,249,525 €, European Union (G.A. 101157493). Role: Task Leader (T4.5 – Deployment and testing of prototype wave attenuator at sea).

6. **DIGI4ECO** – DIGItal Twin-sustained 4D ECOlogical monitoring of restoration in fishery-depleted areas. Mar 2024 – Feb 2028. PI: Dr. Aguzzi (ICM-CSIC). Funding: 9,489,597.50 €, European Union (G.A. 101112883-GAP-101112883). Role: Task Leader (T7.1 – Preparation of demo mission trials).

7. **REDRESS** – REstoration of Deep-sea habitats to Rebuild European SeaS. Feb 2024 – Jan 2028. PI: Dr. Danovaro (UPM). Funding: 8,829,382 €, European Union (G.A. 101135492-GAP-101135492). Role: Researcher.

8. **BluE** – Bioinspired Legged Underwater Robotics for Ecological Monitoring and Exploration. Sep 2023 – Aug 2025. PI: Dr. Aguzzi (ICM-CSIC). Funding: 165,000 €, European Union MSCA Postdoctoral Fellowship (G.A. 101061354). Role: Proponent and Grantee.

9. **PLOME** – Plataforma de Larga Duración para la Observación de los Ecosistemas Marinos. Jan 2023 – Dec 2024. PI: Dr. Aguzzi (ICM-CSIC). Funding: 190,000 €, Ministerio de Ciencia e Innovación (PLEC2021-007525/AEI/10.13039/501100011033). Role: Researcher.

10. **BITER-LANDER** – Esfuerzo conjunto entre BIología y TEcnología para monitorear y recuperar especies y ecosistemas impactados por la pesca. Jan 2023 – Dec 2023. PIs: Dr. Aguzzi (ICM-CSIC), Dr. del Río (UPC). Funding: 167,197 €, Ministerio de Ciencia e Innovación (PID2020-114732RB-C32). Role: Researcher.

11. **Soft Grip.** Jan 2021 – Dec 2022. PI: Dr. Cianchetti (SSSA). Funding: ~3 M€, EU Horizon 2020 (G.A. 101017054). Role: Technical Manager and Researcher.

12. **Hybrid Heart.** Jan 2021 – Apr 2021. PI: Dr. Kluin (Academic Medical Center). Funding: ~3 M€, EU Horizon 2020 (G.A. 767195). Role: Researcher.

13. **EELS** – Exobiology Extant Life Surveyor. Apr 2019 – Jun 2019. PI: Dr. Carpenter (NASA JPL). Funding: NASA JPL Internal Research Program. Role: Researcher.

14. **GOLD** – Guardian of the Ocean Legged Drone. May 2019 – Jul 2021. PI: Dr. Calisti (SSSA). Funding: ~30,000 USD, National Geographic Society (G.A. NGS-56544T-19). Role: Researcher and Project Manager (since Jan 2021).

15. **SILVER** – Seabed Interaction Legged Vehicle for Exploration and Research. Jun 2017 – Oct 2017. PI: Dr. Calisti (SSSA). Funding: ~5,000 USD, National Geographic Society (G.A. CP-056ET-17). Role: Researcher.

16. **Pasteur2** – Training platform for laparoscopic surgery. Sep 2012 – Feb 2013. PI: Dr. Mohamedally (UCL). Funding: UCL Software Engineering Project. Role: Microcontroller programming.

CONTRACTS

1. **Underwater Drone for Arctic Ecosystem Restoration.** Jan 2025 – Dec 2027. PI: Dr. Marcello Calisti (Scuola Superiore Sant'Anna, Italy). Funding body: Real Ice Development Company Ltd. Funding: 945,000 €. Role: Control Engineer and Researcher.

2. **RoSC** – Robotics for the Automatic Sorting of Crustaceans. Apr 2022 – Aug 2022- PI: Dr. Elena Piana, Noola Redclaw (UK). Funding body: Small Business Research Initiative (SBRI–UKRI). Funding: ~50,000 £. Role: Control Engineer and Researcher.

3. **Blue Resolution.** Jul 2018 – Jun 2021. PI: Dr. Marcello Calisti (Scuola Superiore Sant'Anna; University of Lincoln since Jan 2021). Funding body: Dario ARBI S.p.A., Funding: ~200,000 €. Role: Researcher (Jul 2018 – Jun 2021); Project Manager (Jan 2021 – Jun 2021).

STUDENT SUPERVISION

PHD

1. Mrs. Idiondo Molina, Aihnoa. Doctorate program in Marine Science and Technology at UPC, Barcelona, Spain. Thesis title: (Provisional) Can robotics and AI assist restoration and monitoring activities across contrasting benthic ecosystems, from shallow seagrass meadows to shelf-edge fishing ground. — March 2026 - Ongoing

Co-directed with Dr. Nixon Bahamon (ICM-CSIC).

2. Mr. Aguirregomezcorta Aina. Doctorate program in Automation, Robotics, and Vision at UPC, Barcelona, Spain. Thesis title: MARL-based Benthic Navigation for Bioinspired Underwater Legged Robotics. Co-directed with Prof. D.M. Toma (UPC). — January 2025 - Ongoing

MASTER

1. Mr. Penna. M.Sc. in Mechanical Engineering at Mines Nancy (École des Mines de Nancy), France. Thesis title: Testing and design of a waterproof leg manipulator for underwater robotics. — September 2025

2. Mrs. Li. M.Sc. in Mechatronics Engineering at University of Trento, Italy. Thesis title: Locomotion Analysis and Bio-inspired Gait Generation for Hexapod Robot. Co-directed with Prof. A. Del Prete. — September 2025

3. Mrs. Greco. M.Sc. in Biomedical Engineering at University of Pisa, Italy. Thesis title: Design and Implementation of a Soft Robotic Arm for Exploring Deep Sea Burrows. Co-directed with Prof. M. Cianchetti. — April 2025

4. Mrs. Loizzo. M.Sc. in Mechatronics Engineering at Polytechnic University of Turin, Italy. Thesis title: Creation of an RL Environment to Monitor Ocean Features with Autonomous Vehicles. Co-directed with Prof. Alessandro Rizzo, Dr. Ivan Masmitja. — April 2025

5. Mr. Fonseca. M.Sc. in Bionics Engineering at Scuola Superiore Sant'Anna (SSSA), Pisa, Italy. Thesis title: Design and assessment of a modular single-legged underwater robot. Co-directed with Prof. L. Ricotti. — October 2021

BACHELOR

1. Mr. Bertran. B.Sc. in in Marine Science and Technology at Universitat Politècnica de Catalunya (UPC), Barcelona, Spain. Supervisor: Dr. J. del Río (UPC). Thesis title: Camera system for ULR SILVER2. — December 2025

2. Mrs. Rigual. B.Sc. in in Marine Science and Technology at Universitat Politècnica de Catalunya (UPC), Barcelona, Spain. Co-directed with Dr. D. M. Toma (UPC). Thesis title: ROS2 Gazebo design and modelling of a robotic arm for Nephrops in situ observation. — December 2025

3. Mr Soler. B.Sc. in in Marine Science and Technology at Universitat Politècnica de Catalunya (UPC), Barcelona, Spain. Co-directed with Dr. D. M. Toma (UPC). Thesis title: Mechanical design for a rolling robot with an eccentric rotating mass. — October 2025

4. Costin Boiciuc. B.Sc. in Electrical and Electronic Engineering at Saxion University of Applied Sciences, Netherlands. Co-directed with Dr. D. M. Toma (UPC). Thesis title: Design and development of an underwater rolling robot for benthic ecosystem exploration. — July 2025

5. Ana Cantero. B.Sc. in Marine Science and Technology at Universitat Politècnica de Catalunya (UPC), Barcelona, Spain. Supervisor: Dr. J. del Río (UPC). Thesis title: Development of a ROS-based robotic simulator and control architecture for a benthic crawler. — July 2023

TEACHING ACTIVITIES

- Two hours seminar on “Legged Robots Underwater” at the ESEAS - Exploring sea and space: technologies, opportunities and challenges”, Scuola Superiore Sant'Anna. — September 2024
- One hour seminar on “Underwater robotic mobile platforms and underwater legged robots” at the 2024 Summer School on Marine Restoration and Digital Twins, University of Gotenburg. — June 2024
- Two hours seminar on “Legged Robots Underwater” at the ESEAS - Exploring sea and space: technologies, opportunities and challenges”, Scuola Superiore Sant'Anna. — September 2023
- Four hours class on “Notes on underwater technologies” within the Course in Fishieries, Master Study in Fisheries and Wildlife Management, University of Zagreb. — May 2023
- Six hours class on “Modelling and Simulation Tools for Industry 4.0” within the professional course Tirrenicaltech — July 2020
- Four hours class on “Design and control of bioinspired legged underwater robots” within the Course in “Human and Animal models in biorobotics”, Master degree in Bionics Engineering, Scuola Superiore Sant’Anna and University of Pisa. — 2019
- Four hours class on “Design and control of bioinspired legged underwater robots” within the Course in “Human and Animal models in biorobotics”, Master degree in Bionics Engineering, Scuola Superiore Sant’Anna and University of Pisa. — 2018

DISSEMINATION AND OUTREACH

EVENTS

- Roundtable on Robotics in the Mediterranean Europe: a real opportunity for development — May 2026
- 2025 Climate Robotic Summit — May 2025
- Seminar “Il futuro del nostro mare”. Italian Naval Academy. Barcelona, Spain. — September 2024
- Science is Wonderful! 2023. Brussels, Belgium. — March 2023
- Bright Night 2021 – European Researchers Night, Livorno, Italy — September 2021
- Biennale di Venezia, Livorno, Italy — September 2021
- MPA - Meloria Protection & Art, Livorno, Italy — June 2021
- Sea Drone Tech Summit, Ostia, Italy — October 2019
- International Robotics week, Pisa, Italy — September 2018
- Open Day at Research Center on Underwater Robotics and Marine Technology, Livorno Italy — May 2018
- Open Day at the BioRobotics Institute, Pontedera, Italy — October 2017
- ERL Emergency Major Tournament, Piombino, Italy — September 2017
- International Robotics week, Pisa, Italy — September 2017

SELECTED MEDIA COVERAGE

- Climate Robotic Summit - online event. — May 2025
- Green Flags – podcast. — May 2023
- Hydro International - online magazine — April 2022
- ARBI Spa - Webinar — February 2021
- Abyss Cleanup - You Tube channel — August 2020
- Tech Explore - online magazine — May 2020
- RAI Super Quark - Italian public television — August 2020

- Costa Ovest Info - online magazine — June 2018
- RAI Linea Verde - Italian public television — March 2018

HIGH SCHOOL PROJECTS

- Mentoring student team from Col·legi El Carme de Lleida, Spain, for the Lego Competition The SUBMERGED Season - FIRST LEGO League an educational STEM Programme for everybody between 4 and 16. Responsible teacher: Sabrina Buscio Olivera. Responsible teacher: Achilleas Kostoulas, Georgios Gidas. — December 2024
- Mentoring student team from Developmental Center of Thessaly (AKETH), Greece, for the Lego Competition The SUBMERGED Season - FIRST LEGO League an educational STEM Programme for everybody between 4 and 16. Responsible teacher: Sabrina Buscio Olivera. — October 2024
- Mentoring five high school students from Liceo Scientifico Federigo Enriques, Livorno, Italy in the framework of the Alternanza Scuola Lavoro project. Responsible teacher: Mila Barbieri — 2017-2018

CONFERENCES AND WORKSHOPS ATTENDANCE

- 2nd SUNBIO Workshop on Posidonia Oceanica Restoration. Oral presentation by Mrs. Idiondo. — June 2026
- ROSCon España 2025, Barcelona. Oral presentation by Mr. Aguirregomezcorta. — November 2025
- OCEANTECH @ IROS2025 Marine Robotics Marine Science Workshop, Hangzhou, China. Poster presentation. — October 2025
- 12th International Workshop on Marine Technology (MARTECH), Pasaia, Spain. Oral presentation. — May 2025
- 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan. Oral and poster presentation. — May 2024
- 2024 UNESCO Ocean Decade Conference, Barcelona, Spain. Poster presentation. — April 2024
- 11th International Workshop on Marine Technology (MARTECH), Palma, Spain. Oral presentation. — June 2024
- 2023 IEEE International Conference on Robotics and Automation (ICRA). Attendance. — May 2023
- 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Online. Oral presentation. — September 2021
- Oceans 2020 MTS/IEEE Conference, Marseille, France. Oral presentation by Prof. Calisti. — June 2020
- Sea Drone Tech Summit, Ostia, Italy. Oral presentation. — October 2019
- 2019 IRIM 3D Event, Rome, Italy. Poster presentation. — October 2019
- 36th Southern California Control Workshop, Los Angeles, USA. Attendance. — May 2019
- 2018 IEEE International Conference on Soft Robotics (RoboSoft). Poster presentation. — April 2018
- Oceans 2015 MTS/IEEE Conference, Genova, Italy. Student staff member. — May 2015

SELECTED

FIELD EXPERIENCE

- Two-week fieldwork in Hailouoto (Finland), within the Underwater Drone for Arctic Ecosystem Restoration contract to test the under-ice vehicle and ice penetration system — February 2026
- Two 10-day fieldwork visits at the Kristineberg Center for Marine Research and Innovation, University of Gothenburg (Sweden), within the DIGIT4ECO and REDRESS EU projects on the deployment of the ROSSIA crawler for benthic ecosystem monitoring. — 2023-2024
- Three 1-week fieldwork workshops at TEAGASC (Dublin, Ireland) within the SoftGrip EU project, on data collection from human mushroom harvesters and the development of an automated harvesting system. — 2022
- One-week fieldwork campaign in Cariati, Italy, within the GOLD project, focused on benthic environment monitoring using the SILVER2 underwater legged robot. — June 2021

ACADEMIC ACTIVITIES

WORKSHOP AND TRAINING EVENTS ORGANIZATION

- AMBRE @IROS 2026. Adaptive Manipulation: Bridging Rigid and Soft Robotics in Underwater Environments Workshop. Pittsburgh, USA — October 2026
- SUNBIO Workshop “Sustainable Wave Energy Harvesting and Marine Habitat Restoration Integrated with Marine Robotic Systems”. Barcelona, Spain — November 2025
- OCEANTECH @IROS 2025. Marine Robotics Marine Science Workshop. Hangzhou, China — October 2025
- DIGI4ECO and MERLIN Summer School on Marine Restorarion and Digital Twin. Kristineberg Research Station, Sweden. — June 2024

INVITED TALKS

- SUNBIO and MERLIN project meetings: “Bioinspired Underwater Legged Robots for Ecological monitoring and Exploration”. Prof. Papaelias (UB, UK). — May 2025
- “Bio-robotic technologies for Marine Science”. Prof. Wu (CUHK, China). — February 2025
- “Robots for Ecological monitoring and Exploration: insights from a bio-inspired underwater robots”. Prof. Calisti (University of Lincoln, UK). — November 2024
- “Bioinspired machines in the field”. Prof. Thomsen (UGOT, Sweden) — June 2023
- “Bioinspired Underwater Legged robots for Ecological Monitoring and Exploration”. Dr. Consensi (NOC, UK). — January 2022
- “A bio-inspired benthic approach to underwater robotics”, Dr. Carpenter (NASA-JPL, USA) — August 2019

VISITING SCHOLAR HOSTING

- Dr. Niccolò Pagliarani (Scuola Superiore Sant’Anna). Hosted at ICM-CSIC within the Erasmus+ Mobility Training Programme — July 2025
- Dr. Lucrezia Lorenzon (Scuola Superiore Sant’Anna). Hosted at ICM-CSIC within the Erasmus+ Mobility Training Programme — July 2024
- Mr. Vasta (CREA-IT). Hosted at ICM-CSIC. — July 2023

EDITING

- Associate Editor ICRA 2024 – Session on Soft Robotics — 2024

- Topic in Frontiers in Robotics and AI: “The preparation and dish up of an English breakfast with robots (PUB.R) competition: cooking and serving as proxy to showcase the next generation of robots”. 2024
- Topic in MDPI - “Advanced Sensors and Robotic Platform Technologies for the Exploration and Monitoring of Marine Ecosystem and far Beyond” 2023

PEER-REVIEWING

Journals: Scientific Reports, IEEE Robotics and Automation Letters, IEEE Robotics and Automation Magazine, IEEE Transaction on Control System Technologies, International Journal of Robotic Research, Soft Robotics, Bioinspiration and Biomimetics, Nature Communications, Sensors, Frontiers in Robotics and Automation. Journal of Marine Science and Engineering, Journal of Chinese Ocean Engineering.
Conferences: IEEE ICRA, IEEE IROS, IEEE BioRob, IEEE RoboSoft

THESIS EVALUATION COMMITTEES

- President, TFM evaluation committee, Universitat Pompeu Fabra. Master in Intelligent Interactive Systems. Candidate Lorenzo Clementi. July 2026

PROFESSIONAL MEMBERSHIPS

- European Robotic Forum (ERF) Topic Group on Marine Robotics
- European Robotic Forum (ERF) Topic Group on Bio-inspired Robotics
- ARIS (Association of Italian Researchers in Spain)

PROFESSIONAL TRAINING

- Rethinking doctoral thesis supervision: best practices to optimize processes and outcomes. ICM-CSIC, Barcelona, Spain November 2025
- Short-Term Joined Staff Training”. 3-day workshop organized within the Action C1 of the Erasmus+ project “Into the Deep”. Vilanova i la Geltrú, Spain. October 2024
- Breaking the surface (BTS) 2018, 10th International Interdisciplinary Field Workshop of Marine Robotics and Applications, Biograd na Moru, Croatia October 2018